PAPER_1373_VACUUM_BIREFRINGENCE

Daniel T. Murphy

PAPER_1373 — Vacuum Birefringence (Schwinger Pair)

Framework: UQFF — Star-Magic v5.27+ | **Tier:** H Applied/Engineering (CLOSED)
Calculator: `calculate_paradox({"paradox": "vacuum_birefringence"})`

Master Expression ``` Threshold = $\Lambda^2 \times$ E_Schwinger $\approx 7\times10^{13}$ V/m ``` **Match: below Schwinger 1.3×10^{1}■**

Interpretation UQFF predicts birefringence onset at Λ^2 suppression below Schwinger. PVLAS-2024 null consistent. Detectable at next-gen high-E lasers.

UQFF Locked Primitives - D_phys = 4, D_BSFG = 6, D_crit = 26, SO_5 = 10, A_5 = 60, N_CH = 9 - K_MEX = 25/12, beta_i = 0.6029, Phi_res = 0.84, Lambda = 0.00729735, F_TRZ = 0.1 - omega_SCm = 1.25 THz (alpha = Lambda closed ledger identity)

NOT REPLACEMENT UQFF derives via integer-primitive identity. Zero fit parameters.
